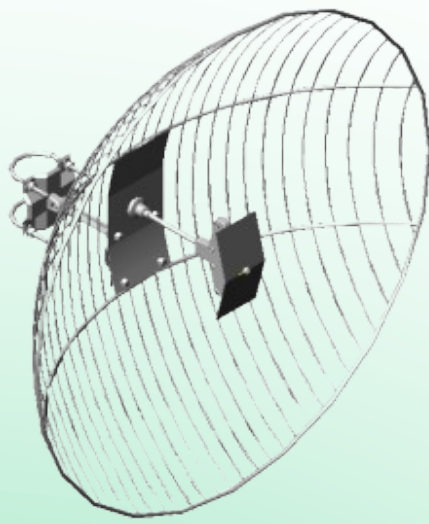




2.4GHz Yagi-fed Parabolic Reflector Antenna Datasheet

A high-gain directional antenna designed for 2.4GHz WiFi applications, providing excellent directivity and reduced power loss for improved wireless connectivity.

Presented by

José Pérez Clar
Ricardo Pereira González

Presented to

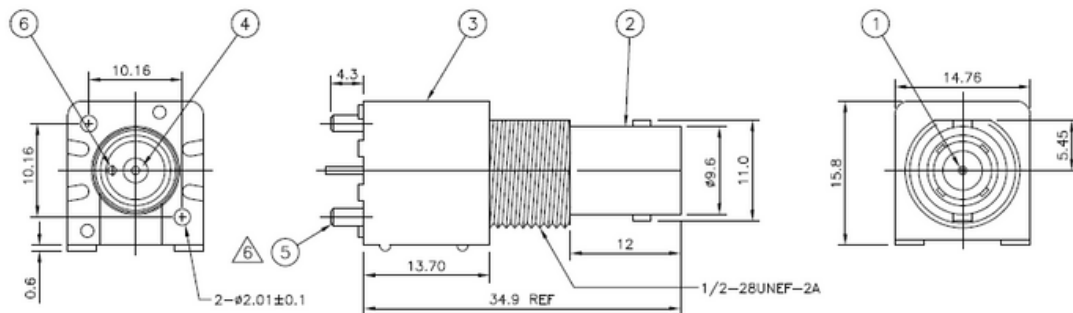
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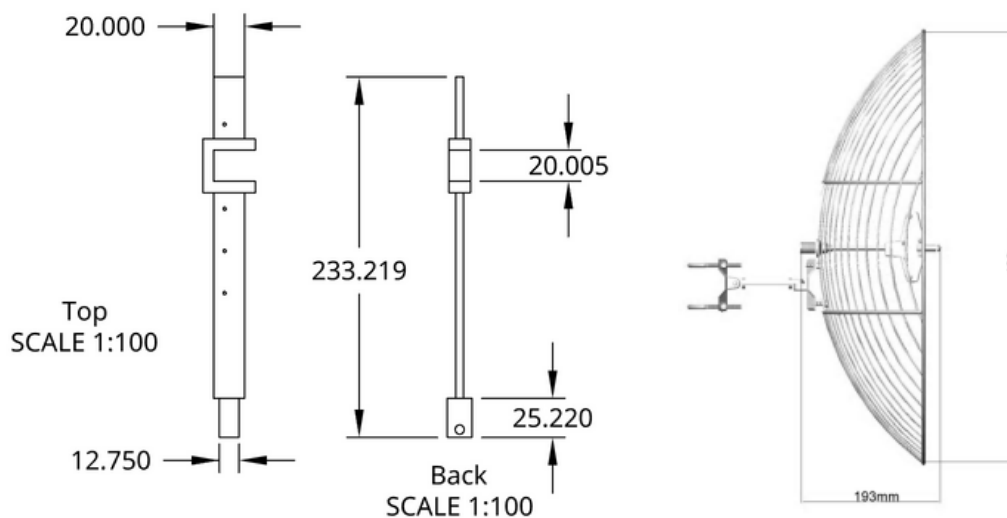
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Dimensions of the antenna

Antenna's Dipole Main Dimensions



Antenna's Supports Main Dimensions



Antenna's Additional Dimensions

- Length Dipole (LA): 60.5mm
- Length Reflector (LR): 453.6mm
- Length Directors (LD): 55.4mm
- Distance from Dipole to Reflector (deltaR): 10.1mm
- Increment of distance from Dipole to each Director (deltaD): 15.1mm

Behavioural Details

Frequency Band: 2.355 GHz to 2.415 GHz

Adaptation Values and Graphs

Return Loss:

- Minimum Return Loss: -20.87 dB
- Maximum Return Loss: -0.54 dB
- Return Loss at 2.4GHz: -19.83 dB

Impedance at 2.4GHz:

- Real Part (R): 42.33 Ω
- Imaginary Part (X): -5.50 Ω

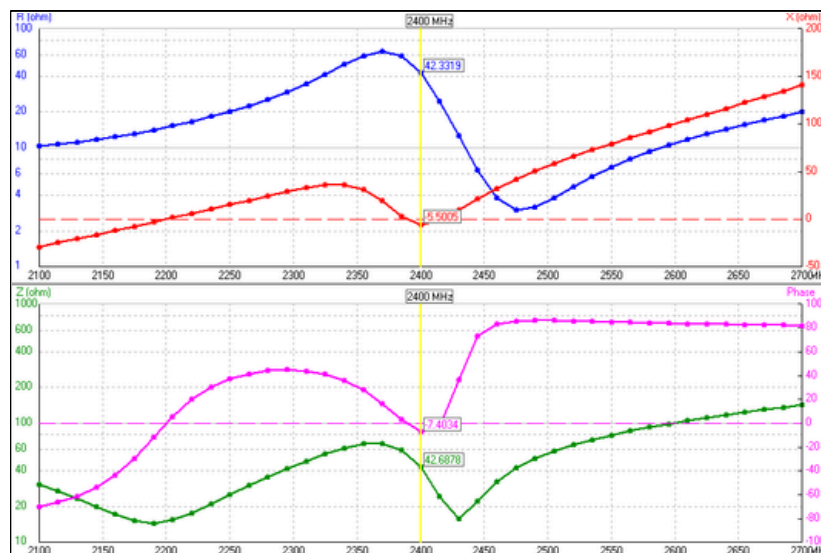


Figure 1: Impedances Graph

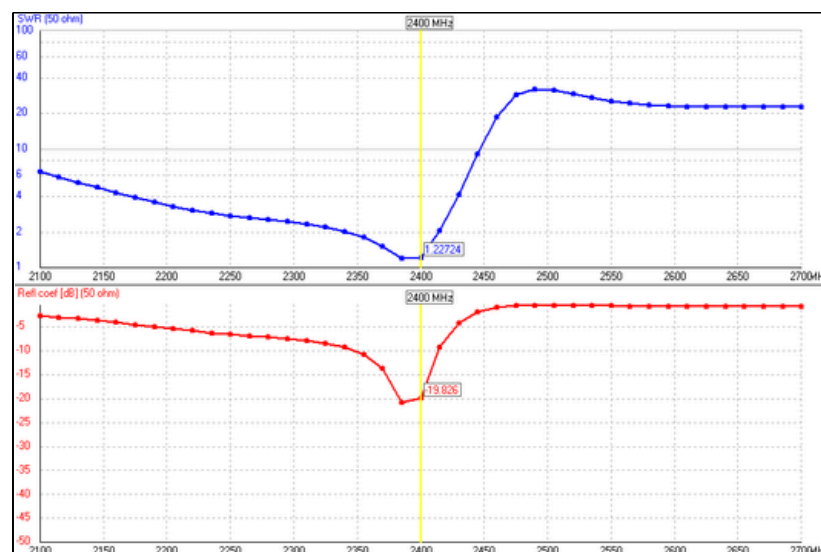


Figure 2: Reflection Coefficient Graph

Behavioural Details

Radiation Data at 2.4GHz

Max Gain: 15.23 dBi

Front-to-Back Ratio (F/B): 42.5 dB

Beamwidth:

- Vertical Plane (Elevation): 54°
- Horizontal Plane (Azimuth): 360°

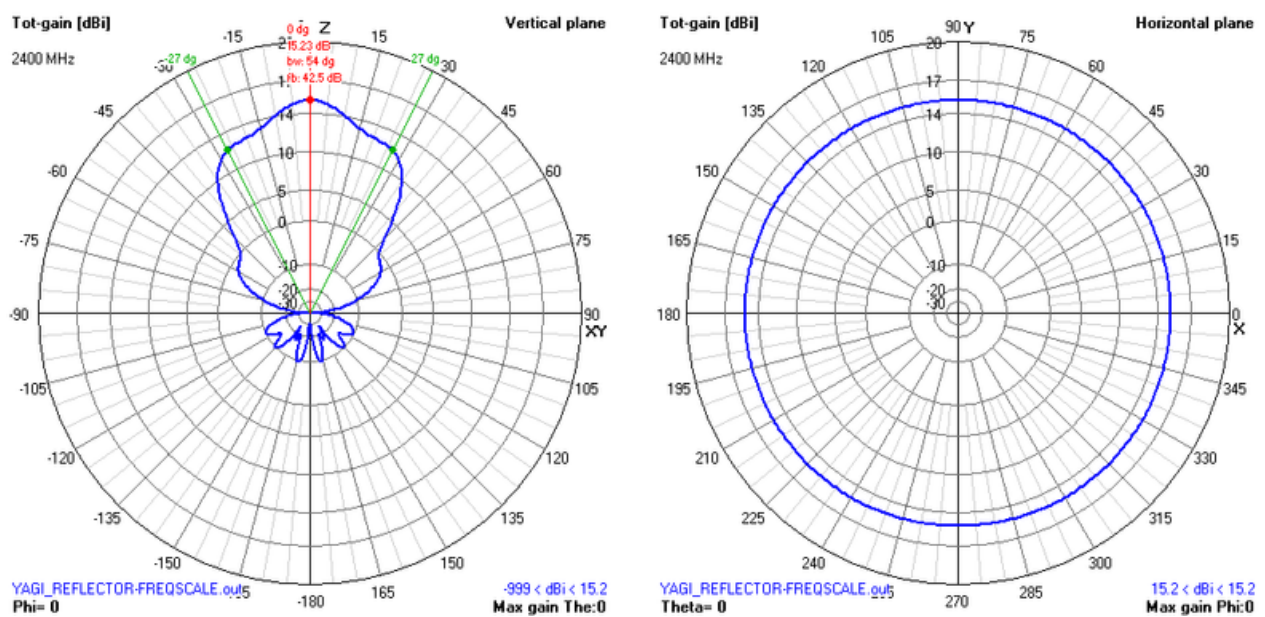


Figure 3: 2D Radiation diagrams

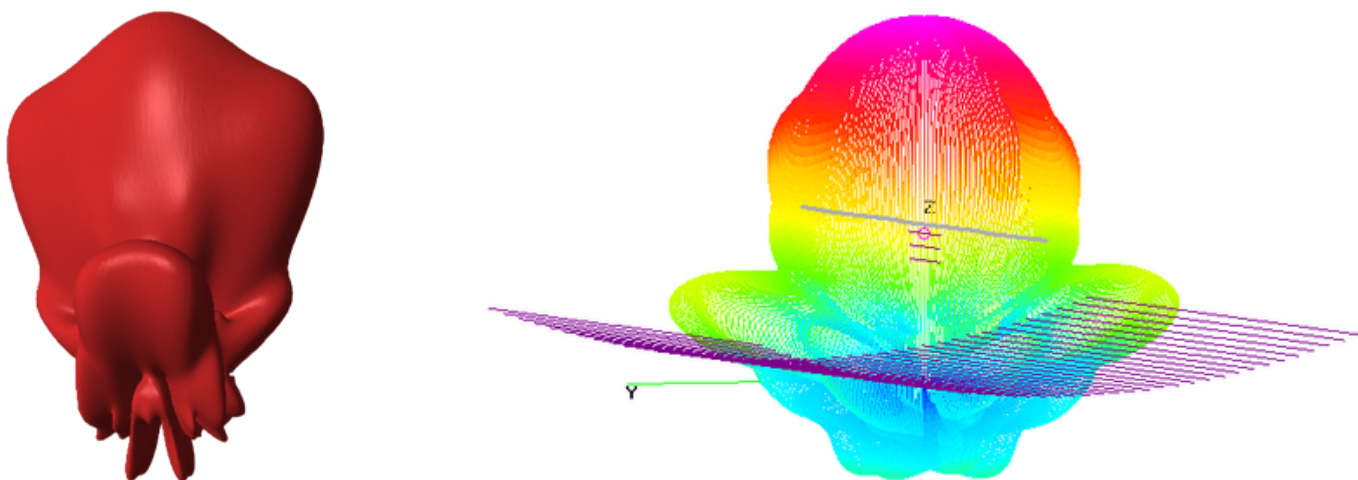


Figure 4: 3D Radiation diagrams



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**Questions?
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